

Sales Performance & Business Insights Report

Student Project Report | Focus: Data Analysis & Power BI

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1. Executive Summary

This project presents an independent sales performance analysis using the commercial transaction dataset powerbi/cleaned_sales_data_powerbi.csv spanning historical sales from January 2003 through May 2005.

The dataset contains:

- 2,823 transaction line items
- 307 unique purchase orders
- \$10,032,628.85 total commercial revenue
- 92 unique corporate customer accounts
- Historical sales from January 2003 through May 2005

The analysis is fully aligned with the finalized Power BI dashboard “Sales Performance Executive Summary” (Page: Executive Summary).

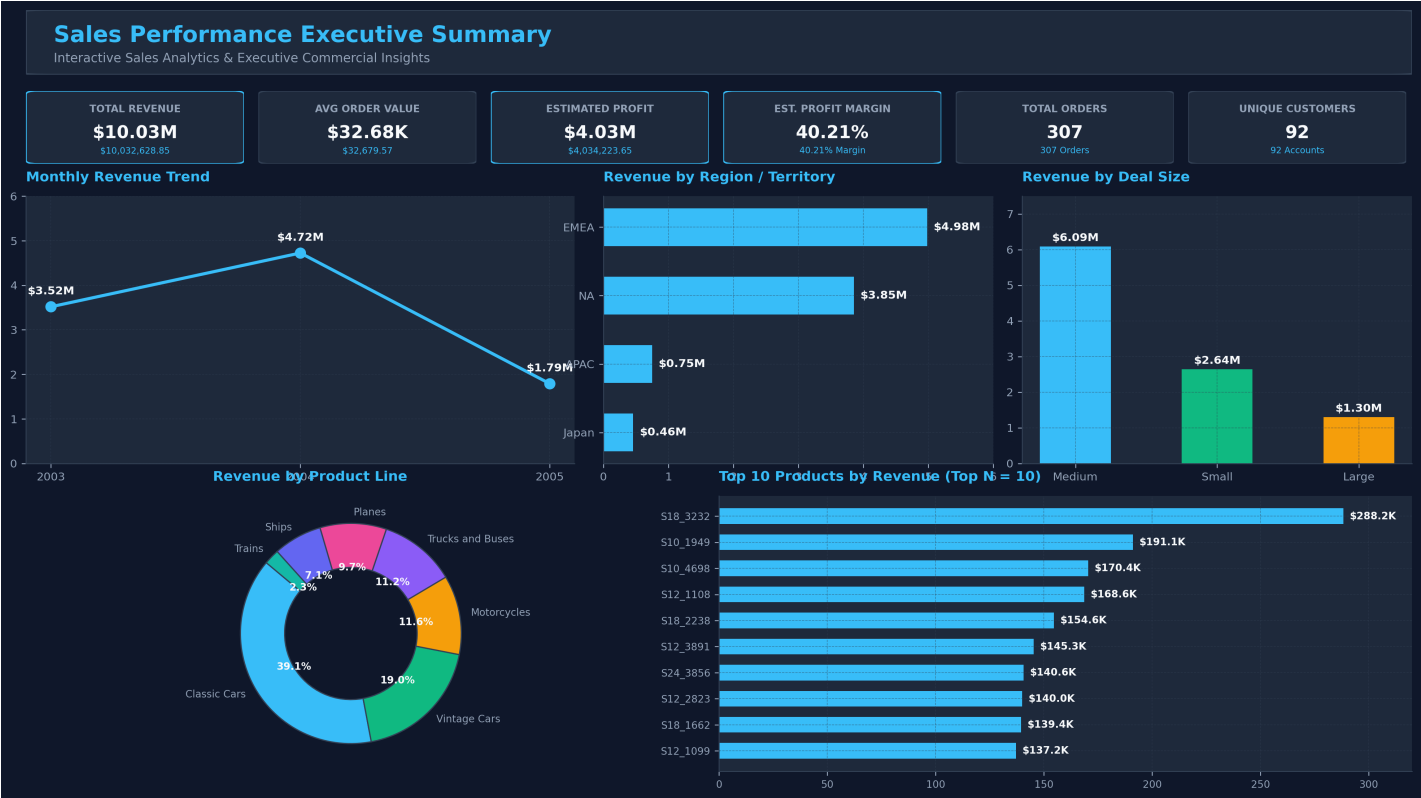


Figure 1. Sales Performance Executive Summary dashboard.

■ **Profit Estimation Methodology Notice:** The source dataset does not contain actual historical accounting cost or net profit figures. Cost of Goods Sold (COGS) is estimated at **60% of MSRP** ($QUANTITYORDERED \times MSRP \times 60\%$). Therefore, all profitability values must be explicitly called **Estimated Profit** and **Estimated Profit Margin**. They should not be interpreted as actual accounting profit.

2. Key Performance Indicators (KPI Summary)

KPI Metric	Unfiltered Value	Metric Explanation / Scope
Total Revenue	\$10,032,628.85	Cumulative sales revenue generated across all 2,823 transaction records.
Average Order Value	\$32,679.57	Average revenue generated per unique purchase order (\$10.03M / 307 orders).
Total Estimated Profit	\$4,034,223.65	Cumulative estimated profit derived from the 60% MSRP cost baseline assumption.
Estimated Profit Margin	40.21%	Overall estimated commercial profit margin percentage across total revenue.
Total Orders	307	Distinct count of purchase orders processed (ORDERNUMBER).
Unique Customers	92	Distinct count of B2B corporate customer accounts (CUSTOMERNAME).

Additional Supporting Metric: **Total Units Sold: 99,067** physical product units delivered across all purchase orders (supporting volume metric).

3. Data-Driven Business Insights

4.1 Revenue Trends Over Time

- **2003 Revenue:** \$3,516,979.54
- **2004 Revenue:** \$4,724,162.60
- **Growth Rate:** Revenue grew by **34.32%** from 2003 to 2004.
- **2005 Observed Revenue:** \$1,791,486.71 (covering transactions from January through May 2005).
- **November Peak Periods:** November 2003 revenue reached \$1,029,837.66 and November 2004 revenue reached \$1,089,048.01.
- **Insight:** November recorded the highest monthly revenue in both 2003 and 2004, standing out as the peak revenue period in the dataset.

4.2 Product Line Performance

1. **Classic Cars:** \$3,919,615.66 (39.07% share) — *Highest revenue-generating product line*
2. **Vintage Cars:** \$1,903,150.84 (18.97% share)
3. **Motorcycles:** \$1,166,388.34 (11.63% share)
4. **Trucks and Buses:** \$1,127,789.84 (11.24% share)
5. **Planes:** \$975,003.57 (9.72% share)
6. **Ships:** \$714,437.13 (7.12% share)
7. **Trains:** \$226,243.47 (2.26% share)

4.3 Regional / Territory Analysis

- **EMEA** (Europe, Middle East & Africa): \$4,979,272.41 (49.63% share) — *Largest territory*
- **North America (NA):** \$3,852,061.39 (38.40% share)
- **APAC** (Asia-Pacific): \$746,121.83 (7.44% share)
- **Japan:** \$455,173.22 (4.54% share)

4.4 Deal Size Performance

- **Medium Deals:** \$6,087,432.24 (60.68% share) — *Contributes the largest share of revenue*
- **Small Deals:** \$2,643,077.35 (26.34% share)
- **Large Deals:** \$1,302,119.26 (12.98% share)

4.5 Top 10 Products by Revenue

1. **S18_3232** — Classic Cars — \$288,245.42 (*Top product SKU*)
2. **S10_1949** — Classic Cars — \$191,073.03
3. **S10_4698** — Motorcycles — \$170,401.07
4. **S12_1108** — Classic Cars — \$168,585.32
5. **S18_2238** — Classic Cars — \$154,623.95
6. **S12_3891** — Classic Cars — \$145,332.04
7. **S24_3856** — Classic Cars — \$140,626.90
8. **S12_2823** — Motorcycles — \$140,006.16
9. **S18_1662** — Planes — \$139,421.97
10. **S12_1099** — Classic Cars — \$137,177.01

4.6 Customer Account Performance

1. **Euro Shopping Channel** — Spain — \$912,294.11 (*Top customer account*)
2. **Mini Gifts Distributors Ltd.** — USA — \$654,858.06
3. **Australian Collectors, Co.** — Australia — \$200,995.41
4. **Muscle Machine Inc** — USA — \$197,736.94
5. **La Rochelle Gifts** — France — \$180,124.90

4.7 Estimated Profitability Performance

- **Total Estimated Profit:** \$4,034,223.65
- **Estimated Profit Margin:** 40.21%
- *Note:* These values are estimates based on the 60% MSRP cost assumption ($\text{QUANTITYORDERED} \times \text{MSRP} \times 60\%$) and do not represent actual accounting profit.

4. Strategic Business Recommendations

1. **Prioritize Core Product Categories:** Focus inventory planning and sales resources on **Classic Cars** (\$3.92M) and **Vintage Cars** (\$1.90M), which together drive over 58% of total commercial revenue.
2. **Support Primary Regional Markets:** Maintain strong sales, marketing, and distribution coverage in **EMEA** (\$4.98M) and **North America (NA)** (\$3.85M), which account for over 88% of global sales.
3. **Incentivize Medium Deal Transitions:** Develop pricing structures, volume discounts, and sales strategies that encourage customer movement from **Small** deal sizes into **Medium** deal tiers (\$6.09M revenue share).
4. **Maintain Top Customer Accounts:** Focus key account management efforts on major corporate accounts such as **Euro Shopping Channel** (\$912.29K) and **Mini Gifts Distributors Ltd.** (\$654.86K) to preserve core revenue streams.
5. **Manage Inventory Around High-Revenue Months:** Prepare inventory stocking, fulfillment capacity, and operational workflows ahead of peak revenue demand periods, particularly **November**.

5. Conclusion

The sales analysis demonstrates strong commercial performance with **\$10.03M** in total revenue and a **40.21% Estimated Profit Margin** based on the defined cost-estimation methodology.

The key findings highlight:

- Total commercial revenue reached **\$10,032,628.85** across 307 orders.
- **Classic Cars** is the leading product line generating **\$3.92M** (39.07% share).
- **EMEA** is the top-performing territory contributing **\$4.98M** (49.63% share).
- **Medium** deal sizes contribute the largest share of revenue at **\$6.09M** (60.68% share).
- **S18_3232** is the top-performing product SKU generating **\$288,245.42**.
- **Euro Shopping Channel** is the leading corporate customer account generating **\$912,294.11**.

Overall, this project demonstrates how data analysis and Power BI visualization can be used to evaluate sales performance, identify business patterns, and develop data-driven recommendations.

Remember that profitability figures in this analysis represent **Estimated Profit** based on the defined cost assumption and should not be interpreted as actual accounting profit.